

Ubuntu DHCP 서버 구축 실습 정리

1. DHCP 패키지 설치

apt install -y isc-dhcp-server 명령어를 사용하여 isc-dhcp-server 패키지를 설치

```
To restore this content, you can run the 'unminimize' command.
Last login: Tue Sep  9 08:49:02 2025 from 172.16.23.1
root@sheep:~# apt install -y isc-dhcp-server
```

2. DHCP 설정 파일 수정

설정 파일 (/etc/dhcp/dhcpd.conf)을 편집기(nano or vim)에서 열어 사진과 같이 net ID와 서브넷, 할당할 ip 범위, 게이트웨이, DNS 서버를 차례로 지정

```
# fixed-address fantasia.example.com;
#}

# You can declare a class of clients and then do address allocation
# based on that.  The example below shows a case where all clients
# in a certain class get addresses on the 10.17.224/24 subnet, and all
# other clients get addresses on the 10.0.29/24 subnet.

#class "foo" {
# match if substring (option vendor-class-identifier, 0, 4) = "SUNW";
#}

#shared-network 224-29 {
# subnet 10.17.224.0 netmask 255.255.255.0 {
#   option routers rtr-224.example.org;
# }
# subnet 10.0.29.0 netmask 255.255.255.0 {
#   option routers rtr-29.example.org;
# }
# pool {
#   allow members of "foo";
#   range 10.17.224.10 10.17.224.250;
# }
# pool {
#   deny members of "foo";
#   range 10.0.29.10 10.0.29.230;
# }
#}
subnet 172.16.23.0 netmask 255.255.255.0 {
    range 172.16.23.20 172.16.23.50;
    option routers 172.16.23.30;
    option domain-name-servers 8.8.8.8;
}

"/etc/dhcp/dhcpd.conf" 118L, 3815B
```

그리고 /etc/default/isc-dhcp-server 파일을 편집기를 이용해서 자신의 이더넷 인터페이스 번호를 적어 넣는다.

```
# On what interfaces should the DHCP server (dhcpcd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens33"
INTERFACESv6=""
```

```
root@sheep:/etc/netplan# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
    link/ether 00:0c:29:fd:d9:24 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 13.3.0.2/24 brd 13.3.0.255 scope global ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:febd:d924/64 scope link
        valid_lft forever preferred_lft forever
```

3. DHCP 서비스 재시작

systemctl restart isc-dhcp-server 명령어를 사용하여 DHCP 서비스를 재시작

```
root@sheep:~# systemctl restart isc-dhcp-server
root@sheep:~# █
```

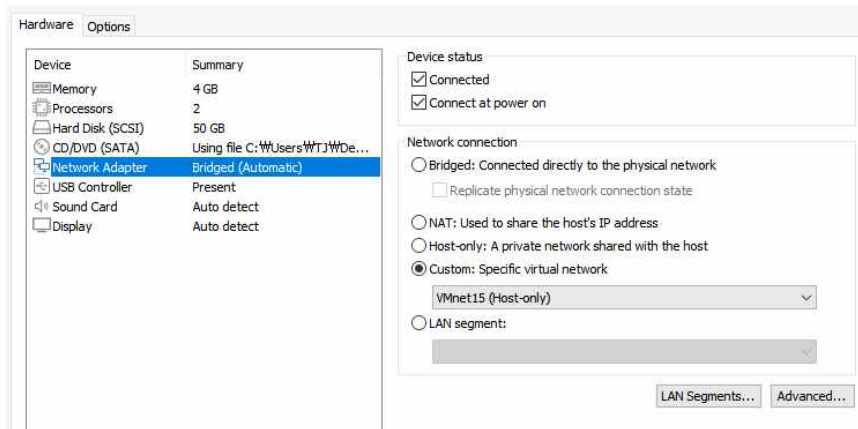
이후 systemctl status isc-dhcp-server 명령어를 통해 DHCP 서버가 active(running) 상태임을 확인

```
root@sheep:~# systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-09-09 09:33:38 UTC; 41s ago
     Docs: man:dhcpd(8)
    Main PID: 4211 (dhcpd)
      Tasks: 1 (limit: 4548)
     Memory: 3.7M (peak: 3.8M)
        CPU: 10ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─4211 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/d

Sep 09 09:33:38 sheep sh[4211]: Wrote 0 leases to leases file.
Sep 09 09:33:38 sheep dhcpd[4211]: PID file: /run/dhcp-server/dhcpd.pid
Sep 09 09:33:38 sheep dhcpd[4211]: Wrote 0 leases to leases file.
Sep 09 09:33:38 sheep dhcpd[4211]: Listening on LPF/ens33/00:0c:29:fd:d9:24/172.16.23.0/24
Sep 09 09:33:38 sheep sh[4211]: Listening on LPF/ens33/00:0c:29:fd:d9:24/172.16.23.0/24
Sep 09 09:33:38 sheep sh[4211]: Sending on LPF/ens33/00:0c:29:fd:d9:24/172.16.23.0/24
Sep 09 09:33:38 sheep sh[4211]: Sending on Socket/fallback/fallback-net
Sep 09 09:33:38 sheep dhcpd[4211]: Sending on LPF/ens33/00:0c:29:fd:d9:24/172.16.23.0/24
Sep 09 09:33:38 sheep dhcpd[4211]: Sending on Socket/fallback/fallback-net
Sep 09 09:33:38 sheep dhcpd[4211]: Server starting service.
lines 1-21/21 (END)
```

5. 네트워크 설정 변경 (VMware)

네트워크 어댑터 모드를 Bridged 에서 Custom (VMnet15, Host-only)로 변경 후



5. VPC 에서 할당받은 IP 확인

ip dhcp 명령어를 통해 vpc 에서 할당받은 ip 확인

```
PC1>
PC1> ip dhcp
DORA IP 172.16.23.20/24 GW 172.16.23.30

PC1> sh ip

NAME          : PC1[1]
IP/MASK       : 172.16.23.20/24
GATEWAY       : 172.16.23.30
DNS           : 8.8.8.8
DHCP SERVER   : 172.16.23.35
DHCP LEASE    : 596, 600/300/525
DOMAIN NAME   : example.org
MAC           : 00:50:79:66:68:00
LPORT        : 10002
RHOST:PORT    : 127.0.0.1:10003
MTU           : 1500

PC1> █
```